

ALEXANDROV-TYPE THEOREM FOR SINGULAR CAPILLARY CMC HYPERSURFACES IN THE HALF-SPACE

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- Capillary CMC hypersurface: variational approach
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- Montiel-Ros's argument
- Main result and sketch of the proof

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- Delgadino-Maggi's approach
- Extension to the capillary case

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Classical Alexandrov's theorem

Theorem (Alexandrov's theorem)

Let Σ be a closed, compact hypersurface embedded in $\mathbf{R}^{n+1}$. If the mean curvature H is constant, then Σ must be a sphere.

Heintze-Karcher inequality: alternative proof of the Alexandrov theorem

Theorem (Heintze-Karcher, Ros, Montiel-Ros)

For a bounded domain Ω in $\mathbf{R}^{n+1}$ with smooth and mean convex boundary $\Sigma = \partial\Omega$, it holds that

$$\int_{\Sigma} \frac{1}{H} \geq \frac{n+1}{n} |\Omega| = \frac{1}{n} \int_{\Sigma} \langle x, \nu \rangle,$$

with equality holds if and only if Σ is a sphere.

Heintze-Karcher inequality: alternative proof of the Alexandrov theorem

Theorem (Heintze-Karcher, Ros, Montiel-Ros)

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with equality holds if and only if Σ is a sphere.

Theorem (Minkowski formula)

For a bounded domain Ω in $\mathbf{R}^{n+1}$ with smooth boundary $\Sigma = \partial\Omega$, there holds

$$\int_{\Sigma} (n - H(x) \langle x, \nu \rangle) dA = 0,$$

where ν denotes the outwards pointing unit normal of $\partial\Omega$ in $\mathbf{R}^{n+1}$.

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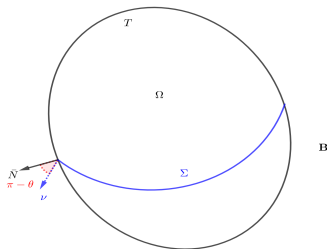
Critical points of the $\mathcal{A}$ -functional

Definition ($\mathcal{A}$ -functional)

Given an open container $\mathbf{B} \subset \mathbf{R}^{n+1}$ with smooth boundary $\partial\mathbf{B}$. For any bounded open set $\Omega \subset \mathbf{B}$ with piecewise smooth boundary $\partial\Omega = \Sigma \cup T$ as follows.

Given $\theta \in (0, \pi)$, $c > 0$, then the $\mathcal{A}$ -functional of Ω is given by

$$\mathcal{A}(\Omega; \mathbf{B}) = |\Sigma| - \cos \theta |T| - c|\Omega|.$$



Critical points of the $\mathcal{A}$ -functional

- We say that Ω is the *critical point of the $\mathcal{A}$ -functional* if

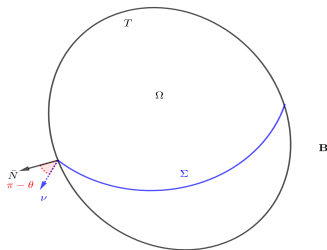
$$\frac{d}{dt} \Big|_{t=0} \mathcal{A}(f_t(\Omega); \mathbf{B}) = 0,$$

where $\{f_t\}_{|t|<\delta}$ is any one-parameter family of C^2 -diffeomorphisms in $\mathbf{R}^{n+1}$ with $f_0 = \text{Id}$, $\text{spt}(f_t - \text{Id})$ is compact in $\mathbf{R}^{n+1}$ and $f_t(\partial\mathbf{B}) \subset \partial\mathbf{B}$.

Critical points of the $\mathcal{A}$ -functional

- **fact:** the $\mathcal{A}$ -stationarity of Ω implies:
 1. Σ is of constant mean curvature and the constant is given by c ;
 2. the Young's law holds, namely, $\langle \nu, \bar{N} \rangle = -\cos \theta$ along the boundary of Σ .

Such hypersurfaces Σ are called *capillary CMC hypersurfaces in $\mathbf{B}$* .



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Alexandrov theorem for capillary CMC hypersurfaces: half-space

When $\mathbf{B} = \mathbf{R}_+^{n+1}$,

- for $\theta \in (0, \pi)$, **Wente 80'**: any embedded capillary hypersurface in the half space whose mean curvature function is symmetric in certain sense ($H = -ku(x_{n+1}) + c$, $u \in C^1$, k somehow measures the gravity), must be axially symmetric, where the axis is vertical to the hyperplane $\text{span}\{x_1, \dots, x_n\}$.

Wente's proof (**moving plane method**) implies the Alexandrov's theorem in the half-space (rigidity of CMC capillary hypersurface).

Alexandrov theorem for capillary CMC hypersurfaces: half-space

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Wente's proof (**moving plane method**) implies the Alexandrov's theorem in the half-space (rigidity of CMC capillary hypersurface).

- when $\theta = \pi/2$ (free boundary): by simply reflecting the hypersurface across the supporting hyperplane! One gets the Heintze-Karcher inequality together with the Alexandrov's theorem.

goal: to establish a Heintze-Karcher-type inequality in the half-space for smooth capillary CMC hypersurface with a generic contact angle $\theta \in (0, \pi)$.

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Sketch of proof: the flow method

Assume that Ω is a bounded open set with smooth, mean convex boundary $\Sigma = \partial\Omega$. Denote by $\{\kappa_i(x)\}_{i=1}^n$ the principal curvatures of Σ at $x \in \Sigma$, indexed in increasing order so that $\kappa_n \geq H/n > 0$. Set $u(y) = \text{dist}(y, \partial\Omega)$ to be the distance function to $\partial\Omega$, which is defined on $\mathbf{R}^{n+1}$.

Define

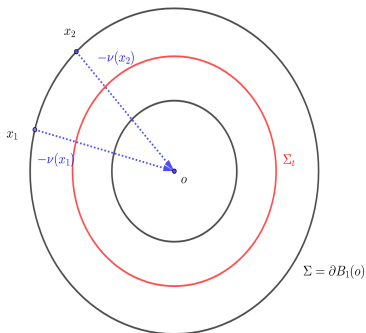
$$Z = \{(x, t) \in \partial\Omega \times \mathbf{R} : 0 < t \leq \frac{1}{\kappa_n(x)}\},$$

and

$$\zeta(x, t) = x - t\nu(x), \quad (x, t) \in Z.$$

Sketch of proof: the flow method

Note that: for the unit round sphere, one has $\kappa_n \equiv 1$ on Σ and hence for any fixed t , $\Sigma_t := \zeta(\Sigma, t)$ is also a round sphere.



Sketch of proof: the flow method

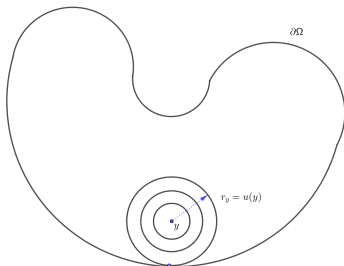
For a general mean convex domain, the Heintze-Karcher inequality is established through the following steps:

Step 1. Surjectivity: $\Omega \subset \zeta(Z)$.

For any $y \in \Omega$, consider the foliation of concentric spheres $\{\partial B_r(y)\}_{r>0}$, suppose that as r increases, the first touching of the foliation with Σ occurs at some x_y for some $r_y > 0$. It is easy to see that

$$u(y) = \text{dist}(y, x_y) = r_y, \quad 1/r_y \geq \kappa_n(x_y), \quad \text{i.e., } u(y) \leq 1/\kappa_n(x_y).$$

In particular, this implies that $y = \zeta((x_y, u(y))) \in \zeta(Z)$.



Sketch of proof: the flow method

Step 2. Applying the area formula.

$$\begin{aligned} |\Omega| &\leq |\zeta(Z)| \leq \int_{\zeta(Z)} \mathcal{H}^0(\zeta^{-1}(y)) dy = \int_Z J^Z \zeta d\mathcal{H}^{n+1} \\ &= \int_{\Sigma} dA \int_0^{1/\kappa_n(x)} \prod_{i=1}^n (1 - t\kappa_i(x)) dt. \end{aligned}$$

Sketch of proof: the flow method

Step 2. Applying the area formula.

$$\begin{aligned}
 |\Omega| &\leq |\zeta(Z)| \leq \int_{\zeta(Z)} \mathcal{H}^0(\zeta^{-1}(y)) dy = \int_Z J^Z \zeta d\mathcal{H}^{n+1} \\
 &= \int_{\Sigma} dA \int_0^{1/\kappa_n(x)} \prod_{i=1}^n (1 - t\kappa_i(x)) dt.
 \end{aligned}$$

By the AM-GM inequality and the fact that $\kappa_n \geq H/n$, we arrive at

$$\begin{aligned}
 |\Omega| &\leq \int_{\Sigma} dA \int_0^{1/\kappa_n(x)} \left(\frac{1}{n} \sum_{i=1}^n (1 - t\kappa_i(x)) \right)^n dt \\
 &\leq \int_{\Sigma} dA \int_0^{n/H} \left(1 - t \frac{H}{n} \right)^n dt = \frac{n}{n+1} \int_{\Sigma} \frac{1}{H} dA.
 \end{aligned}$$

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Main result: Heintze-Karcher-type inequality

Our Heintze-Karcher-type inequality for capillary hypersurfaces in the half-space is as follows:

Theorem (Jia-Wang-Xia-Z. 22')

Let $\theta_0 \in (0, \pi)$ and let $\Sigma \subset \overline{\mathbf{R}_+^{n+1}}$ be a smooth, compact, embedded, strictly mean convex hypersurface that meets $\partial\mathbf{R}_+^{n+1}$ with contact angle $\theta(x) \leq \theta_0$ for every $x \in \partial\Sigma$. Let Ω denote the enclosed domain by Σ and $\partial\mathbf{R}_+^{n+1}$. Then it holds

$$\int_{\Sigma} \frac{1 - \cos \theta_0 \langle \nu, E_{n+1} \rangle}{H} dA \geq \frac{n+1}{n} |\Omega|. \quad (1)$$

Equality in (1) holds if and only if Σ is a spherical cap that meets $\partial\mathbf{R}_+^{n+1}$ with constant contact angle θ_0 .

Main result: Heintze-Karcher-type inequality

Proposition (Minkowski-type formula, Ainouz-Souam 16')

Let $\theta_0 \in (0, \pi)$ and $\Sigma \subset \overline{\mathbf{R}_+^{n+1}}$ be a smooth, compact, embedded hypersurface that meets $\partial\mathbf{R}_+^{n+1}$ with a constant contact angle θ_0 . Then it holds that

$$\int_{\Sigma} n(1 - \cos \theta \langle \nu, E_{n+1} \rangle) - H \langle x, \nu \rangle dA = 0.$$

Main result: Heintze-Karcher-type inequality

Proposition (Minkowski-type formula, Ainouz-Souam 16')

Let $\theta_0 \in (0, \pi)$ and $\Sigma \subset \overline{\mathbf{R}_+^{n+1}}$ be a smooth, compact, embedded hypersurface that meets $\partial\mathbf{R}_+^{n+1}$ with a constant contact angle θ_0 . Then it holds that

$$\int_{\Sigma} n(1 - \cos \theta \langle \nu, E_{n+1} \rangle) - H \langle x, \nu \rangle dA = 0.$$

On the other hand, this together with the Heintze-Karcher inequality, provides an alternative proof of the Alexandrov's theorem.

Theorem

Let $\theta_0 \in (0, \pi)$ and $\Sigma \subset \overline{\mathbf{R}_+^{n+1}}$ be a smooth, compact, embedded hypersurface that meets $\partial\mathbf{R}_+^{n+1}$ with a constant contact angle θ_0 . If H is a constant, then Σ must be a spherical cap that meets $\partial\mathbf{R}_+^{n+1}$ with constant contact angle θ_0 .

Sketch of proof: finding suitable flow

To adapt the Montiel-Ros's argument to our setting, we consider the following flow: given $\theta_0 \in (0, \pi)$.

Define

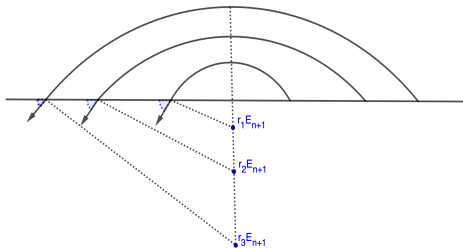
$$Z = \{(x, t) \in \Sigma \times \mathbf{R} : 0 < t \leq \frac{1}{\kappa_n(x)}\},$$

and

$$\begin{aligned}\zeta_{\theta_0} : Z &\rightarrow \mathbb{R}^{n+1}, \\ \zeta_{\theta_0}(x, t) &= x - t(\nu(x) - \cos \theta_0 E_{n+1})\end{aligned}$$

Sketch of proof: finding suitable flow

- Note that for the unit θ_0 -cap (spherical cap that meets $\partial\mathbf{R}_+^{n+1}$ with the constant contact angle θ_0 , e.g. $\partial B_1(-\cos\theta_0 E_{n+1}) \cap \overline{\mathbf{R}_+^{n+1}}$), one has $\kappa_n \equiv 1$ on Σ , and for any fixed t , $\Sigma_t := \zeta_{\theta_0}(\Sigma, t)$ is also a θ_0 -cap.



Sketch of proof: finding suitable flow

- A simple computation shows that for any $(x, t) \in Z$,

$$J^Z \zeta_{\theta_0}(x, t) = (1 - \cos \theta_0 \langle \nu, E_{n+1} \rangle) \prod_{i=1}^n (1 - t \kappa_i).$$

It follows that If Σ is mean convex, then

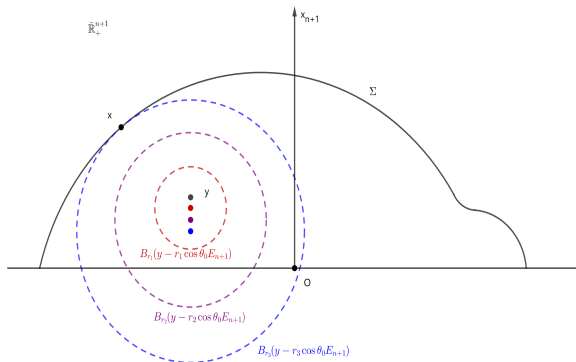
$$\int_Z J^Z \zeta_{\theta_0} d\mathcal{H}^{n+1} = \int_{\Sigma} (1 - \cos \theta_0 \langle \nu, E_{n+1} \rangle) dA \int_0^{1/\kappa_n(x)} \prod_{i=1}^n (1 - t \kappa_i) dt.$$

By the AM-GM inequality,

$$\begin{aligned} \int_Z J^Z \zeta_{\theta_0} d\mathcal{H}^{n+1} &\leq \int_{\Sigma} (1 - \cos \theta_0 \langle \nu, E_{n+1} \rangle) dA \int_0^{\frac{n}{H(x)}} \left(1 - t \frac{H(x)}{n}\right)^n dt \\ &= \frac{n}{n+1} \int_{\Sigma} \frac{(1 - \cos \theta_0 \langle \nu, E_{n+1} \rangle)}{H} dA. \end{aligned}$$

Sketch of proof: Surjectivity of the flow, a new foliation

For any $y \in \Omega$, consider the foliation of spheres with shifted centers $\{\partial B_r(y - r \cos \theta_0 E_{n+1})\}_{r>0}$, suppose that as r increases, the first touching of the foliation with Σ occurs at some x_y for some $r_y > 0$. It can be proved that $x_y \notin \partial \Sigma$ by using the contact angle condition $\theta(x_y) \leq \theta_0$. Therefore $1/r_y \geq \kappa_n(x_y)$, which implies that $y = \zeta_{\theta_0}((x_y, r_y)) \in \zeta_{\theta_0}(Z)$.



Conclusion of the proof

By surjectivity,

$$|\Omega| \leq |\zeta_{\theta_0}(Z)| \leq \frac{n}{n+1} \int_{\Sigma} \frac{(1 - \cos \theta_0 \langle \nu, E_{n+1} \rangle)}{H} dA.$$

The equality takes if and only if Σ is a θ_0 -cap.

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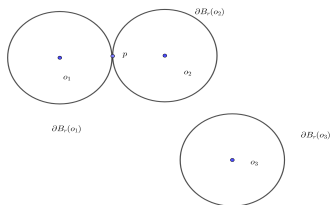
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Delgadino-Maggi's approach: main result

The classical Alexandrov's theorem for closed hypersurface in $\mathbf{R}^{n+1}$ can be extended to the context of sets of finite perimeter as follows:

Theorem (Delgadino-Maggi 19')

Among sets of finite perimeter and finite volume, finite unions of balls with equal radii are the only critical points of the Euclidean isoperimetric problem.



Delgadino-Maggi's approach: main result

Their theorem can be stated in a rather direct manner as follows:

For a closed hypersurface $\Sigma \subset \mathbf{R}^{n+1}$, let $\text{reg}\Sigma$ denote the regular part of Σ , defined as: for any $x \in \text{reg}\Sigma$, there exists a neighborhood such that Σ is C^2 in this neighborhood. Let $\text{sing}\Sigma := \Sigma \setminus \text{reg}\Sigma$ denote the singular part of Σ , which is relatively closed in Σ by definition.

Moreover, let Ω denoted the bounded enclosed region of Σ and let ν denote the outer unit normal along $\text{reg}\Sigma$.

Delgadino-Maggi's result: For a closed hypersurface $\Sigma \subset \mathbf{R}^{n+1}$, if

- 1 $\mathcal{H}^n(\text{sing}\Sigma) = 0$;
- 2 it satisfies the distributional structure: for some constant $c > 0$,

$$\int_{\Sigma} \text{dis}_{\Sigma} X d\mathcal{H}^n = \int_{\Sigma} \langle X, c\nu \rangle d\mathcal{H}^n, \quad \forall X \in C_c^1(\mathbf{R}^{n+1}; \mathbf{R}^{n+1}),$$

then Σ must be a finite union of spheres with equal radii.

Sketch of proof

Their proof is based on an adaption of the Montiel-Ros's argument to sets of finite perimeter.

Compared to the classical Montiel-Ros's argument, it is natural to consider

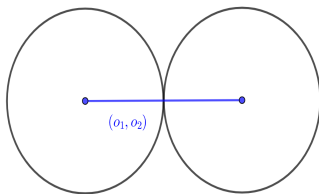
$$Z = \{(x, t) \in \text{reg}\Sigma \times \mathbf{R} : 0 < t \leq \frac{1}{\kappa_n(x)}\},$$

and

$$\zeta(x, t) = x - t\nu(x), \quad (x, t) \in Z.$$

Sketch of proof

However, it is not in general true that $\zeta(Z)$ is still surjective, even for the target that we wish to prove.



Sketch of proof

Delgadino-Maggi proved that $|\zeta(Z) \setminus \Omega| = 0$ by first constructing a large subset $\Omega^* \subset \Omega$ and then show that

1 $|\Omega \setminus \Omega^*| = 0,$

2 $|\zeta(Z) \setminus \Omega^*| = 0.$

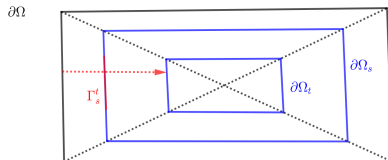
- Recall that $u(y) = \text{dist}(y, \partial\Omega)$ is defined to be the distance function from $\partial\Omega$.

Roughly speaking, we define Ω^* to be the set of points in Ω such that the distance function is differentiable. From the Rademacher's theorem one easily sees that $|\Omega \setminus \Omega^*| = 0$ holds.

Formal definition of $\Omega^\star$

For any $0 < s < t < \infty$, let $\Omega_s = \{y \in \Omega : u(y) > s\}$ be the super level-set of the distance function u .

- Define Γ_s^t = the set of points in Ω such that, for any $y \in \Gamma_s^t$, there exists a unit-speed geodesic $\gamma_y : [0, t] \rightarrow \mathbf{R}^{n+1}$ with $\gamma_y(s) = y$, $u(\gamma_y(r)) = r$ for any $r \in [0, t]$.
- Define $\Gamma_s^+ = \bigcup_{t>s} \Gamma_s^t$.
- Define $\Omega^\star = \bigcup_{s>0} \Gamma_s^+$.



Sketch of proof

- The second step $|\zeta(Z) \setminus \Omega^*| = 0$ is much more complicated, the proof is based largely on the distributional structure:

$$\int_{\Sigma} \text{dis}_{\Sigma} X d\mathcal{H}^n = \int_{\Sigma} \langle X, c\nu \rangle d\mathcal{H}^n, \quad \forall X \in C_c^1(\mathbf{R}^{n+1}; \mathbf{R}^{n+1}).$$

With the distributional structure, we may consider the blow-up process at any $x \in \Sigma$, if $x \in \text{reg}\Sigma$, the distributional structure then becomes:

$$\int_{\tilde{\Sigma}} \text{dis}_{\tilde{\Sigma}} X d\mathcal{H}^n = 0, \quad \forall X \in C_c^1(\mathbf{R}^{n+1}; \mathbf{R}^{n+1}).$$

Similarly, for those $x \in \text{sing}\Sigma$, we may obtain something that is stationary (a stationary rectifiable cone) as the blow-up limit.

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Main result

Let $\overline{\mathbf{R}_+^{n+1}}$ denote the closed half-space. We consider bounded, relatively open set $\Omega \subset \overline{\mathbf{R}_+^{n+1}}$, and denote by $\partial_{\text{rel}}\Omega = \overline{\partial\Omega} \cap \mathbf{R}_+^{n+1}$ the relative boundary.

Theorem (Xia-Z. 23')

Let $\Omega \subset \overline{\mathbf{R}_+^{n+1}}$ be a bounded, relatively open set, if

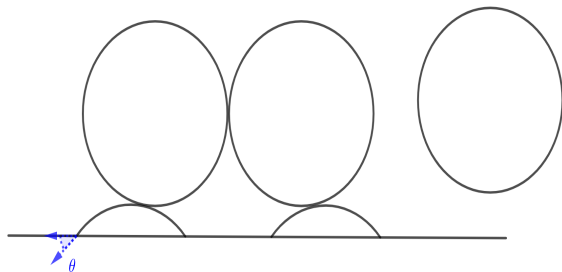
- 1 $\mathcal{H}^{n-1}(\text{sing}\partial_{\text{rel}}\Omega) = 0$, $\partial_{\text{rel}}\Omega \cap \partial\mathbf{R}_+^{n+1}$ is a smooth hypersurface in $\partial\mathbf{R}_+^{n+1}$;
- 2 it satisfies the distributional structure: for $\theta \in (0, \pi)$ and some constant $c > 0$,

$$\begin{aligned} & \int_{\partial_{\text{rel}}\Omega} \text{div}_{\partial\Omega} X d\mathcal{H}^n - \cos\theta \int_{\partial\Omega \cap \partial\mathbf{R}_+^{n+1}} \text{div}_{\partial\mathbf{R}_+^{n+1}} X d\mathcal{H}^n \\ &= \int_{\partial_{\text{rel}}\Omega} \langle X, c\nu \rangle d\mathcal{H}^n - \sin\theta \int_{\partial_{\text{rel}}\Omega \cap \partial\mathbf{R}_+^{n+1}} \langle X, E_{n+1} \rangle d\mathcal{H}^{n-1}, \end{aligned}$$

for any C^1 -vector field X that has compact support,
then $\partial_{\text{rel}}\Omega$ must be a finite union of θ -caps and spheres (that lies completely in $\mathbf{R}_+^{n+1}$) with equal radii.

Main result

Example:



Sketch of proof: shifted distance function

As already discussed, it is natural to define

$$Z = \{(x, t) \in \operatorname{reg} \partial_{\operatorname{rel}} \Omega \times \mathbf{R} : 0 < t \leq \frac{1}{\kappa_n(x)}\},$$

and

$$\zeta_\theta(x, t) = x - t(\nu(x) - \cos \theta E_{n+1}), \quad (x, t) \in Z.$$

But it is in general not true that $\Omega \subset \zeta_\theta(Z)$. Indeed, we have to construct a large subset Ω^* with the properties that

- 1 $|\Omega \setminus \Omega^*| = 0;$

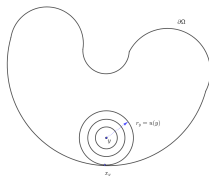
- 2 $|\zeta_\theta(Z) \setminus \Omega^*| = 0.$

To construct Ω^* , a suitable choice of the distance function is needed.

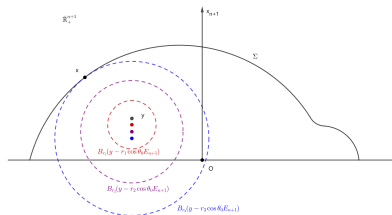
Sketch of proof: shifted distance function - unique point projections

In Delgadino-Maggi's approach, the distance function $u(y)$ can be understood as

$$u(y) = \sup_{r \geq 0} \{r : B_r(y) \cap \partial\Omega = \emptyset\}.$$



(a) unique point projection.



(b) shifted unique point projection

Sketch of proof: shifted distance function

This motivates the following definition.

- Given a bounded, relatively open set $\Omega \subset \overline{\mathbf{R}_+^{n+1}}$ and $\theta \in (0, \pi)$. Let δ_θ be the *shifted distance function with respect to $\partial_{\text{rel}}\Omega$ and θ* , defined as

$$\delta_\theta(y) = \sup_{r \geq 0} \{r : B_r(y - r \cos \theta E_{n+1}) \cap \partial_{\text{rel}}\Omega = \emptyset\}.$$

- By collecting the points of differentiability of δ_θ in $\Omega \cup (\partial\Omega \cap \partial\mathbf{R}_+^{n+1})$, we obtain the desired set Ω^* .

Sketch of proof

Likewise to Delgadino-Maggi's approach, the second step, to show that $|\zeta_\theta(Z) \setminus \Omega| = 0$, is way more complicated. The distributional structure:

$$\begin{aligned} & \int_{\partial_{\text{rel}} \Omega} \operatorname{div}_{\partial \Omega} X \, d\mathcal{H}^n - \cos \theta \int_{\partial \Omega \cap \partial \mathbf{R}_+^{n+1}} \operatorname{div}_{\partial \mathbf{R}_+^{n+1}} X \, d\mathcal{H}^n \\ &= \int_{\partial_{\text{rel}} \Omega} \langle X, \nu \rangle \, d\mathcal{H}^n - \sin \theta \int_{\partial_{\text{rel}} \Omega \cap \partial \mathbf{R}_+^{n+1}} \langle X, E_{n+1} \rangle \, d\mathcal{H}^{n-1}, \end{aligned}$$

for any C^1 -vector field X that has compact support, is essentially used in this step.

Thank you for your attention!